

Derivation of the recursive formula for the Exponential Moving Average (EMA)

First – order linear ordinary differential equations :

$$\dot{\omega}_m(t) + \frac{1}{T_m} \omega_m(t) = \frac{1}{T_m} \omega_{sp}(t)$$

Let the initial time be 0, and use the general solution formula :

$$\omega_m(t) = e^{-\frac{t}{T_m}} \omega_m(0) + \int_0^t e^{-\frac{t-\tau}{T_m}} \frac{1}{T_m} \omega_{sp}(\tau) d\tau$$

Consider the interval from time t_k to time t_{k+1} :

$$\omega_m(t_{k+1}) = e^{-\frac{t_{k+1}-t_k}{T_m}} \omega_m(t_k) + \int_{t_k}^{t_{k+1}} e^{-\frac{t_{k+1}-\tau}{T_m}} \frac{1}{T_m} \omega_{sp}(\tau) d\tau$$

Let $\Delta t = t_{k+1} - t_k$:

$$\omega_m(t_{k+1}) = e^{-\frac{\Delta t}{T_m}} \omega_m(t_k) + \int_{t_k}^{t_{k+1}} e^{-\frac{t_{k+1}-\tau}{T_m}} \frac{1}{T_m} \omega_{sp}(\tau) d\tau :$$

Zero-order hold assumption: $\omega_{sp}(\tau) = \omega_{sp}(t_k), \tau \in [t_k, t_{k+1})$

So,

$$\omega_m(t_{k+1}) = e^{-\frac{\Delta t}{T_m}} \omega_m(t_k) + \omega_{sp}(t_k) \int_{t_k}^{t_{k+1}} e^{-\frac{t_{k+1}-\tau}{T_m}} \frac{1}{T_m} d\tau$$

Let $s = t_{k+1} - \tau$, and $d\tau = -ds$, so :

$$\begin{aligned} \int_{t_k}^{t_{k+1}} e^{-\frac{t_{k+1}-\tau}{T_m}} \frac{1}{T_m} d\tau &= \int_{\Delta t}^0 e^{-\frac{s}{T_m}} \frac{1}{T_m} (-ds) = \int_0^{\Delta t} e^{-\frac{s}{T_m}} \frac{1}{T_m} ds \\ \int_0^{\Delta t} e^{-\frac{s}{T_m}} \frac{1}{T_m} ds &= \left[-e^{-\frac{s}{T_m}} \right]_0^{\Delta t} = -e^{-\frac{\Delta t}{T_m}} + 1 = 1 - e^{-\frac{\Delta t}{T_m}} \end{aligned}$$

Finally,

$$\omega_m(t_{k+1}) = e^{-\frac{\Delta t}{T_m}} \omega_m(t_k) + \omega_{sp}(t_k) \left(1 - e^{-\frac{\Delta t}{T_m}} \right)$$

$$\omega_{k+1} = \alpha \omega_k + (1 - \alpha) \omega_{sp,k}$$

$$\text{There, } \alpha := e^{-\frac{\Delta t}{T_m}}.$$

To find $\hat{T}_m$

*Defining the conditional likelihood of our dataset D
by using an isotropic Gaussian observer model :*

$$D = \{(A_k, b_k)\}_{k=1}^N, b_k = A_k x + \epsilon_k, \epsilon_k \sim \mathcal{N}(0, \sigma^2 I)$$

$$p(\mathcal{D} | T_m) = \prod_{k=1}^N p(b_k | A_k)$$

$$p(b_k | A_k) = \frac{1}{(2\pi)^{3/2} \sigma^3} \exp\left(-\frac{1}{2\sigma^2} \|b_k - A_k x\|^2\right)$$

By Bayes' formula :

$$\begin{aligned} p(T_m | \mathcal{D}) &= \frac{p(\mathcal{D} | T_m) \cdot p(T_m)}{p(\mathcal{D})}, p(T_m) = C_1, p(\mathcal{D}) = C_2 \\ &= \frac{C_1}{C_2} \prod_{k=1}^N \left[\frac{1}{(2\pi)^{3/2} \sigma^3} \exp\left(-\frac{1}{2\sigma^2} \|b_k - A_k x\|^2\right) \right] \end{aligned}$$

$$\begin{aligned} \log p(T_m | \mathcal{D}) &= \log\left(\frac{C_1}{C_2}\right) + \sum_{k=1}^N \left[\log\left(\frac{1}{(2\pi)^{3/2} \sigma^3}\right) - \frac{1}{2\sigma^2} \|b_k - A_k x\|^2 \right] \\ &= \log\left(\frac{C_1}{C_2}\right) + N \log\left(\frac{1}{(2\pi)^{3/2} \sigma^3}\right) - \frac{1}{2\sigma^2} \sum_{k=1}^N \|b_k - A_k x\|^2 \\ &= C_3 - \frac{1}{2\sigma^2} \sum_{k=1}^N \|b_k - A_k x\|^2 \end{aligned}$$

Finally,

$$\begin{aligned} \hat{T}_m &= \arg \max_{T_m} \log p(T_m | \mathcal{D}) \\ &= \arg \max_{T_m} \left(C_3 - \frac{1}{2\sigma^2} \sum_{k=1}^N \|b_k - A_k x\|^2 \right) \\ &= \arg \min_{T_m} \sum_{k=1}^N \|b_k - A_k x\|^2 \end{aligned}$$

Identify the $\mathbf{J}$ and K_{τ_i}

Rotational dynamics equations :

$$\mathbf{J}\dot{\boldsymbol{\omega}}_b - (\mathbf{J}\boldsymbol{\omega}_b) \times \boldsymbol{\omega}_b - \sum_{i=1}^4 \mathbf{r}_{\tau_i} K_{\tau_i} f_i = \sum_{i=1}^4 (\mathbf{r}_{p_i} \times \mathbf{r}_{f_i}) f_i$$

$$\begin{aligned} \mathbf{J}\dot{\boldsymbol{\omega}}_b - (\mathbf{J}\boldsymbol{\omega}_b) \times \boldsymbol{\omega}_b &= \begin{bmatrix} I_{xx}\dot{\omega}_x - I_{yy}\omega_y\omega_z + I_{zz}\omega_z\omega_y \\ I_{xx}\omega_x\omega_z + I_{yy}\dot{\omega}_y - I_{zz}\omega_z\omega_x \\ -I_{xx}\omega_x\omega_y + I_{yy}\omega_y\omega_x + I_{zz}\dot{\omega}_z \end{bmatrix} \\ &= A_J \cdot \begin{bmatrix} I_{xx} \\ I_{yy} \\ I_{zz} \end{bmatrix} \end{aligned}$$

$$\text{with : } A_J := \begin{bmatrix} \dot{\omega}_x & -\omega_y\omega_z & \omega_z\omega_y \\ \omega_x\omega_z & \dot{\omega}_y & -\omega_z\omega_x \\ -\omega_x\omega_y & \omega_y\omega_x & \dot{\omega}_z \end{bmatrix}$$

$$\text{Let } \mathbf{x} := \begin{bmatrix} I_{xx} & I_{yy} & I_{zz} & K_{\tau_1} & K_{\tau_2} & K_{\tau_3} & K_{\tau_4} \end{bmatrix}^\top$$

Finally,

$$A\mathbf{x} = b$$

with :

$$A_{K_\tau} := \begin{bmatrix} -\mathbf{r}_{\tau_1} f_1 & -\mathbf{r}_{\tau_2} f_2 & -\mathbf{r}_{\tau_3} f_3 & -\mathbf{r}_{\tau_4} f_4 \end{bmatrix}$$

$$b := \sum_{i=1}^4 (\mathbf{r}_{p_i} \times \mathbf{r}_{f_i}) f_i$$

$$A := \begin{bmatrix} A_J & A_{K_\tau} \end{bmatrix}$$

Because of $\mathbf{r}_{\tau_i} = [0, 0, \pm 1]^\top$, the first two rows of A_{K_τ} are 0, meaning K_τ contributes nothing to roll and pitch.

Furthermore, in actual flight, the diagonal term of A_J is significantly greater than the cross-reference, so, $\omega_x\omega_y \approx 0, \omega_x\omega_z \approx 0, \omega_y\omega_z \approx 0$

therefore I_{xx} and I_{yy} can be approximated by the following equations :

$$\begin{bmatrix} \dot{\omega}_x & 0 \\ 0 & \dot{\omega}_y \end{bmatrix} \cdot \begin{bmatrix} I_{xx} \\ I_{yy} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

Taking the third row of the equation, at this point :

$$\dot{\omega}_z I_{zz} - \sum_{i=1}^4 K_{\tau} \mathbf{r}_{\tau_i} f_i = 0 \text{ (b的第三行为0)}$$

Assuming the torque coefficients of the four motors are the same, i.e., $K_{\tau_i} = K_{\tau}$

$$\dot{\omega}_z I_{zz} = K_{\tau} \sum_{i=1}^4 \mathbf{r}_{\tau_i} f_i$$

$$\dot{\omega}_z \cdot \frac{I_{zz}}{K_{\tau}} = \sum_{i=1}^4 \mathbf{r}_{\tau_i} f_i$$

So,

$$\mathbf{Ax} = \mathbf{b}$$

$$\text{with : } A := \dot{\omega}_z, x := \frac{I_{zz}}{K_{\tau}}, b := \sum_{i=1}^4 \mathbf{r}_{\tau_i} f_i$$

K_{τ} 对俯仰（Pitch）与横滚（Roll）无作用，
同时在小角速度的假设下， $\dot{\omega}_b \gg \omega_b$ ，令耦合项 ≈ 0

$$\sum_{i=1}^4 \mathbf{r}_{\tau_i} f_i K_{\tau} = \dot{\omega}_z \cdot I_{zz}$$

$$A := \sum_{i=1}^4 \mathbf{r}_{\tau_i} f_i, b := \dot{\omega}_z \cdot I_{zz}, x := K_{\tau}$$

再看等式的第三行，假设四个电机的推力系数相同，即：

$$K_{\tau_i} = K_{\tau}$$

在小角速度的假设下，等式变为：

理论上， $\frac{I_{zz}}{K_{\tau}}$ 的比值已足以精准仿真无人机动力学，
但为了提升可解释性，我们希望将其拆分为转动惯量 I_{zz} 与力矩系数 K_{τ} 两项。

$$A:=\begin{bmatrix} \dot{\omega}_x & -\omega_y\omega_z & \omega_z\omega_y & 0 & 0 & 0 & 0 \\ \omega_x\omega_z & \dot{\omega}_y & -\omega_z\omega_x & 0 & 0 & 0 & 0 \\ -\omega_x\omega_y & \omega_y\omega_x & \dot{\omega}_z & f_1 & -f_2 & f_3 & -f_4 \end{bmatrix}$$

$$b:=\sum_{i=1}^4(\mathbf{r}_{p_i}\times\mathbf{r}_{f_i})f_i$$

$$\mathbf{x}:=\begin{bmatrix} I_{xx} & I_{yy} & I_{zz} & K_{\tau_1} & K_{\tau_2} & K_{\tau_3} & K_{\tau_4} \end{bmatrix}^\top$$